

A Rotavirus-Induced Mouse Model to Study Biliary Atresia and Neonatal Cholestasis.

PMID: 31016660 | Indexed in PubMed

Biliary atresia is a devastating neonatal cholangiopathy that affects both extra- and intrahepatic bile ducts progressing to fibrosis and end-stage liver disease by 2 years of age. Despite re-establishment of biliary drainage following a Kasai portoenterostomy (surgical procedure), many infants develop fibrosis requiring liver transplant. In the murine model of biliary atresia, rhesus rotavirus infection of newborn pups results in a cholangiopathy paralleling human biliary atresia and is used to study mechanistic aspects of the disease. The infected mice displayed histopathological signs similar to human biliary atresia, with bile duct obstruction, bile duct proliferation, and liver inflammation with fibrosis.

[Diagnostic Imaging of Biliary Atresia].

PMID: 36276203 | Indexed in PubMed

Biliary atresia is a rare but significant cause of neonatal cholestasis. An early and accurate diagnosis is important for proper management and prognosis. To diagnose biliary atresia, various imaging studies using ultrasonography, MRI, hepatobiliary scans, and cholangiography can be performed, although ultrasonography is more important for initial imaging studies. In this article, we review the findings of biliary atresia from various imaging modalities, including ultrasonography, MRI, hepatobiliary scans, and cholangiography. The known key imaging features include abnormal gallbladder size and shape, periportal thickening visible as a 'triangular cord' sign, invisible common bile duct, increased hepatic arterial flow, and combined anomalies. Aside from the imaging findings of biliary atresia, we also reviewed the diagnostic difficulty in the early neonatal period and the role of imaging in predicting hepatic fibrosis. We hope that this review will aid in the diagnosis of biliary atresia.

Biliary atresia.

PMID: 2653529 | Indexed in PubMed

Biliary atresia remains a serious and devastating surgical condition of early childhood. Despite portoenterostomy almost half of these children remain jaundiced and suffer repeated attacks of bacterial cholangitis or variceal haemorrhage. Recently liver transplantation offers fresh hope for these children.

Biliary atresia: A comprehensive review.

PMID: 27346637 | Indexed in PubMed

Biliary atresia presents as an obliterative cholangiopathy with neonatal jaundice and pale stools. The disease exhibits aetiological heterogeneity with a multiplicity of potential causative factors, both developmental and environmental. A number of clinical variants making up a minority of all cases can be defined relatively precisely which match suggested aetiology better although in most it still remains speculative. These include the syndromic form (BASM), the cystic form and those associated with CMV IgM antibodies. We review not only the clinical evidence for a developmental or an immune-mediated

aetiology perhaps triggered by perinatal viral exposure but also several other recently suggested concepts such as microchimerism, gene susceptibility and environmental toxins.

Course of life into adulthood of patients with biliary atresia: the achievement of developmental milestones in a nationwide cohort.

PMID: 22626493 | Indexed in PubMed

To investigate the course of life of young adults diagnosed with biliary atresia (BA) in infancy by comparing patients who did and did not underwent transplantation with an age-matched Dutch reference group. All patients from the Dutch BA registry, aged >18 years, were invited to complete the course of life questionnaire. Forty patients participated (response = 74%). Twenty-five had not undergone transplantation; 15 had undergone orthotopic liver transplantation. One significant between-group difference was found, namely in substance use and gambling. BA patients who underwent transplantation reported less use than the reference group ($p = .01$, moderate effect size). Additional moderate effect sizes were found for differences in psychosexual and social development and antisocial behavior. Patients who underwent transplantation had lower scores than one or both other groups. Development of BA survivors who did not undergo transplantation seems not delayed, whereas that of transplanted patients does seem somewhat delayed. However, patients who underwent transplantation display less risk behavior. Larger samples are necessary to confirm these findings.

Diagnostic Performance of Transient Elastography in Biliary Atresia Among Infants With Cholestasis.

PMID: 34027275 | Indexed in PubMed

Biliary atresia (BA) is a challenging liver disease in infancy. Early diagnosis of BA is important for timely hepatportoenterostomy. We evaluated the age-specific diagnostic performance of transient elastography (TE) with a liver stiffness measurement (LSM) greater than 7.7 kPa in BA among infants with cholestasis. A total of 61 infants with cholestasis (5-121 days of age) were enrolled in this prospective follow-up study; 15 infants were BA. Four age groups were defined (≤ 30 , 31-60, 61-90, and 91-180 days). Picrosirius red staining was performed to quantify the percentage of collagen fibers in liver specimens. The utility of an LSM greater than 7.7 kPa for diagnosis of BA among infants with cholestasis was compared among age groups. In all four groups, TE showed high diagnostic power for BA using the criterion of an LSM greater than 7.7 kPa. Positive predictive values were 100%, 100%, and 100% in the groups aged 30 days or younger, 31 to 60 days, and 61 to 90 days, respectively. Respective negative predictive values were 90.9%, 94.7%, and 100%, and respective diagnostic accuracies were 92.9%, 95.2%, and 100%. The positive predictive value, negative predictive value, and diagnostic accuracy were 100%, 100%, and 100%, respectively, for LSM greater than 8.8 kPa in the group aged 91 to 180 days. The LSM was positively correlated with the percentage of collagen fibers stained by picrosirius red ($P = 0.03$). Conclusion: In this prospective follow-up study, TE had good diagnostic accuracy for differentiation of BA from non-BA cholestasis in infants with cholestasis who were 90 days of age or younger. The LSM was significantly positive correlated with the liver fibrosis status stained by picrosirius red in infants with cholestasis.

Cholestasis in Infancy.

PMID: 27975109 | Indexed in PubMed

Jaundice is a key manifestation of hepatobiliary disease in all age groups. Jaundice is a common finding in the first 2 weeks after birth, occurring in 2.4% to 15% of newborns. The neonatal liver is at increased susceptibility to cholestasis, with an incidence ranging from 1 in 2,500 to 1 in 5,000 live births. Etiologies vary, but the most common is biliary atresia. In 2004, the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition published guidelines for the evaluation of cholestasis that clearly stated any infant with jaundice persisting beyond age 2 weeks (3 weeks in breast-fed infants with an otherwise normal history and physical examination) should be evaluated with a fractionated serum bilirubin level. Prompt evaluation, diagnosis, and intervention are vital to optimize timely intervention and improve clinical outcomes. This article discusses the etiology, diagnosis and evaluation of cholestasis in infants. [Pediatr Ann. 2016;45(12):e414-e419.].

Has Rotavirus Vaccination Decreased the Prevalence of Biliary Atresia?

PMID: 30222646 | Indexed in PubMed

Biliary atresia (BA) is a rare neonatal liver disease that causes cholestasis and is the leading indication for pediatric liver transplantation. Although the exact etiology of BA remains unknown, evidence from murine models supports the role of rotavirus infection in the development of BA. In 2006, universal rotavirus vaccination was implemented in the United States. The goal of this study was to determine if the prevalence of BA correlated with the number of annual rotavirus infections. We utilized data from the 1997 to 2012 Kids' Inpatient Database and the 1988 to 2015 Organ Procurement and Transplantation Network to determine the annual number of infant discharges with a primary diagnosis of BA and the number of infants with BA who received a liver transplant, respectively. We obtained the number of annual rotavirus infections from the National Respiratory and Enteric Virus Surveillance System and examined whether trends existed between the data from these 3 sources over time. From 1997 to 2006, the number of positive rotavirus antigen tests remained steady, however a rapid decrease was observed from 2006 to 2012 (8774 to 1277), coinciding with the uptake of rotavirus immunizations nationwide. The number of BA discharges doubled from 1997 to 2003 and again increased from 2006 to 2012 (67 to 137 and 117 to 156), while the number of liver transplants for BA changed very little from 1997 to 2012. The recent implementation of rotavirus vaccination has not had any substantial influence on the prevalence of BA in the United States.

Insignificant seasonal and geographical variation in incidence of biliary atresia in Japan: a regional survey of over 20 years.

PMID: 18082714 | Indexed in PubMed

Biliary atresia (BA) is the leading cause of obstructive jaundice in the newborn and the major indication for liver transplantation in Japan. Viral infection has been implicated in its etiology because of seasonality and space clustering. However, this has been controversial among whites. The present study investigates space-time clustering of the incidence of BA in Japan. Birth prevalence rates of BA were analyzed in infants born in Hyogo prefecture between 1985 and 2004 to assess time clustering among 3 seasons. The

birth prevalence rates were also analyzed for space clustering between the northern (Hokkaido) and the southern islands (Kyushu) (difference of latitude, 10 degrees) based on the Japanese Biliary Atresia Society Survey 1996-2004. We compared the prevalence rates between these groups using relative risks (RRs) calculated from 2 x 2 contingency tables. One hundred nineteen infants were born with BA in Hyogo (1.1/10000 live births). Seasonal clustering in April to July did not significantly differ from that of the reference period (RR, 1.45; 95% confidence interval, 0.93-2.27). Based on Japanese registry data, space clustering between the northern and southern islands did not significantly differ (RR, 1.24; 95% confidence interval, 0.83-1.86). Neither seasonal nor spatial clustering was statistically proven in Japan.

Newborn screening for biliary atresia in the United States.

PMID: 28983658 | Indexed in PubMed

Despite advances in our understanding of the pathogenesis of biliary atresia (BA), BA remains the most common cause of end-stage liver disease in children and the leading indication for pediatric liver transplantation. Age at time of Kasai portoenterostomy (KPE), performed to provide bile drainage, strongly correlates with transplant-free survival, mostly due to progression of intrahepatic fibrosis to cirrhosis. Unfortunately, challenges remain in recognizing that a jaundiced infant may have BA. To better diagnose infants with BA at an earlier age, population-based screening programs in countries such as Taiwan, Japan, and China have utilized stool color cards. Early results have been promising demonstrating earlier diagnosis, earlier KPE, and, hence, improved outcomes. Cost-effectiveness studies focused on stool color card screening in North America where the incidence of BA is much lower also project improved transplant-free survival rate with a savings in terms of healthcare expenditure. There is also evidence that postnatal serum bilirubin levels may also be effective as a screening tool given that all infants with BA exhibit hyperbilirubinemia at birth. The American Academy of Pediatrics (AAP) recently advocated studying the implementation of newborn screening for BA in the United States. Further efforts and analyses within the United States are ongoing, but current evidence is supportive of screening for BA even in low incidence countries.
